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Moving toward belonging in a physics department: What changes are possible with and without a paradigm shift?

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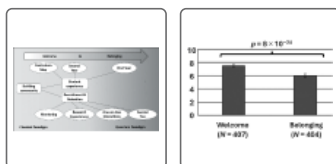
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Abstract

Belonging in a discipline or department is an important aspect of people's persistence in that discipline or department, especially for undergraduate students. Previous research has shown that increasing student feelings of belonging in a department benefits all students and is even more valuable for women, racialized people, and other underrepresented groups. However, helping students to develop feelings of belonging can be difficult and complex. This article presents data from a larger, primarily qualitative case study that was conducted at a research-based university in Canada. It examines how a group of faculty, staff, and students worked toward building community and cultivating belonging in their physics department. The aspects of this process that the professor and student participants experienced, and the challenges and rewards encountered, are described and explained here. Some of the strategies that professor participants undertook were relatively easily accomplished using what we describe here as a "classical" paradigm; the analogy is to the traditional physics culture. One example of a strategy employed under the classical paradigm is hosting a mentoring program for physics majors. Other strategies were quite challenging for many professor participants and required a paradigm shift toward more progressive views, which, following the analogy, we call a "quantum" paradigm. An example is believing and behaving as though students with B grades belong and can succeed in physics. We suggest that the paradigm shift from classical (traditional thinking) to quantum (progressive) is an important barrier that members of the physics community must recognize and overcome to fully address student participation and experiences in physics, although some changes were still possible without making the paradigm shift. The study found a new difference between feeling welcome and feeling a sense of belonging among students, as well as a similar difference among faculty members. We present possible steps that physics faculty and staff may consider along the road to recruitment and retention in physics, which are notable for departments and instructors interested in improving student outcomes, departmental sustainability, and equity, diversity, and inclusion.



Physics Subject Headings (*PhySH*)

[Diversity & inclusion](#)
[Learning environment](#)
[Faculty](#)
[Lower undergraduate students](#)
[Upper undergraduate students](#)

Article Text

I. INTRODUCTION

Belonging means the degree to which one feels valued, secure, and supported within a group [1–3]. Improving student feelings of belonging in the physics community can have many positive effects, such as higher grades [4], greater persistence in degree programs [5–8], greater self-efficacy [9], and improved physical and psychological well-being [1]. By cultivating belonging, we aim to include everyone who would like to participate in the physics community, which would build diversity and long-term sustainability in our discipline. This type of change has long eluded us in physics, while similar disciplines have made substantial recent improvements [10–13]. Reasons for our lack of progress include that many physics faculty may not know how they can increase students' sense of belonging [9,14], they may believe that they do not influence student persistence at all [7], or they may not be comfortable implementing strategies presented in the literature without support [15,16].

This paper presents results from part of a larger case study [17] conducted in one university physics department in Canada where a small group of faculty and staff attempted to increase undergraduate persistence, diversity, and engagement. The focus of this work is on how both faculty and student participants described their thoughts and experiences related to who belongs and how belonging can be achieved in a physics department.

Participants in this study included faculty, staff, and students in a physics department, and the intent was to investigate and improve upon student feelings of belonging as a means of improving retention in the department's undergraduate programs. Interestingly, however, some findings emerged about the faculty and staff perceptions of their own belonging in physics. Therefore, the results include discussions of belonging that relate to both student and professor experiences.

The study focused on how student success and belonging could be improved upon at the departmental level because departments are an appropriate unit for attempting such changes [15]. Course-level success and belonging are also valuable and were encouraged as part of this work, but they were secondary to the overall goal of the project, which was to increase student retention, diversity, and engagement in the department.

II. THEORETICAL FRAMEWORK

Much recent research in physics education has emphasized the importance of building a sense of belonging in physics communities [4, 13,18–20] along with bolstering people's physics identities [3, 9, 21]. These two concepts are strongly interrelated, and both exist precariously within a messy, complex ecosystem [22] that remains harsh, especially for women [23–24], racialized people [25–27], and other underrepresented groups [28]. Members of these groups often leave or avoid physics because they do not feel they belong in the field, for a variety of

societal, circumstantial, and contextual reasons.

A. Physics identity, self-efficacy, and belonging

Physics identity has been described as the feeling of being a “physics person” [29]. Persistence in physics graduate programs and careers is affected by physics identity [5, 29], and gender and race gaps in physics identity have existed for many years [27,29]. There are many influences on physics identity, including self-efficacy and recognition by peers and authority figures [29]. People’s physics identities are constructed over many years, starting long before they arrive on university campuses. This creates a daunting challenge for anyone hoping to improve student feelings of belonging in physics: it involves changing people’s deeply held beliefs about themselves. Improving students’ sense of belonging has been shown to positively affect both the persistent gender gap [1, 9, 30,31] and the racial imbalance [25,26,32,33] in physics.

1. Faculty effects on belonging

University physics faculty can greatly influence physics identity through implicit and explicit recognition of students as capable of success in physics [3, 4, 12], although they may not realize that they hold this power [7]. Recognition of student efforts and abilities [12,21,34] and creating supportive learning environments [9, 19] are the two most important methods by which physics professors, instructors, and teaching assistants can build student physics identities and belonging. But crucially, both of these methods require an underlying belief that, as educators, we can and do influence student experiences of physics.

Faculty beliefs about teaching and supportive departmental cultures are known to have important effects on student belonging [7, 16,35,36]. Unfortunately, the traditional physics culture implicitly discourages student belonging and supportive teaching.

B. The classical paradigm

Introductory physics courses have long been known as “gatekeeper” courses [37]; that is, they are designed and carried out with the goal to “weed out” large numbers of students [38], in what most physics community members incorrectly believe is a meritocratic system [39,23]. This makes sense in a culture of elitism. The underlying belief that physics is only for the best and brightest students aligns with the practice that most introductory physics courses are still being taught traditionally, using lectures as the sole type of instruction [16,39,40] and assuming that those who are talented enough will succeed without support. Importantly, the stereotype that physics is only for “brilliant” people is widely held in society, not only by physicists [10,11]. Combined with the societal association of brilliance with maleness and whiteness [11,41], this creates a large and unspoken barrier to broadening participation in physics.

The *classical paradigm* is the term we use in this study for the set of beliefs about how traditionally taught courses are necessary in physics to maintain the field's elite status. This implies restricting progress in physics courses to only the top tier of students. The classical paradigm is based on the assumption that using traditional teaching methods ensures that only the most capable students will be allowed to persist in physics and that those who discover that they are capable will naturally choose to remain without further action from anyone in the physics community. By definition, then, the classical paradigm creates an elitist culture and contributes greatly to low enrolment.

C. The quantum paradigm

In her 1990 book [42], Sheila Tobias investigated elitism in the academic cultures of physics and chemistry. She wrote,

We need to enlarge what has hitherto been considered the natural pool of recruits to science and be willing to offer new kinds of students a welcome and a chance for success. ... We are obliged to think and think hard not just about who does science and why, but who doesn't do science, and why not. To this end, we should examine, as a beginning construct, the student on the 'second tier'.

The term "second tier" refers to students who are capable and interested in physics and chemistry but were pushed out of the fields during their undergraduate years due to elitist cultures where only "top tier" students were welcomed. This excludes many students who might contribute greatly to the field but whose grades fall just short of the highest tier, or who do not consider themselves brilliant. Many of these students identify as women and/or as part of other underrepresented groups [10–12].

In undergraduate physics, students learn and adopt a shift in how the universe functions when they advance from classical physics to quantum physics. This is usually difficult, but it results in a wider and more accurate perspective of how the universe works. Similarly, we argue in this paper that some physics educators undergo a shift in pedagogical understanding from the classical paradigm described above to the quantum paradigm, which is a set of beliefs about how inclusive teaching and a culture of belonging can be used in physics to improve outcomes for the majority of students while still ensuring that they learn the same material and are appropriately challenged. The quantum paradigm requires a belief that many, if not all, students can succeed in physics, and that, to do so, they need and deserve our support.

Many previous scholars have pointed out that elitism and meritocracy are the source of the physics enrolment and diversity problems [23,43–45]. Some have discussed this as a cultural problem [19,31,42,43,45,46]. Here, we build on these understandings to discuss the problem as a thought experiment: the need to shift from the classical paradigm, including its elitism and the resulting low enrollments, to the quantum paradigm, based on promoting belonging and broader participation.

D. Welcoming vs belonging

Belonging is well defined in the literature as a sense that one's presence is valued and that one is safe and secure [1–3]. Welcoming has not yet been as clearly defined and is only mentioned tangentially in the literature as of this writing. The meaning of “welcome” in this paper is “to greet hospitably and with courtesy or cordiality” or “to accept with pleasure the occurrence or presence of” [47]. Therefore, welcoming people into a community is an initial step toward helping people feel that they belong. That is, welcoming is on the continuum towards belonging. ... You can only belong through other human beings. ... People who only feel welcomed leave, and people who belong stay [48].

The distinction between welcome and belonging is new to the physics education literature. For the purposes of this research, we assume that the two terms are interpreted by participants and readers according to their commonly understood definitions as outlined above.

Much research is extant on physics students' identities and sense of belonging, but the same is not true for physics professors. The only information currently available for professors is about how they can build their students' physics identities. No references were found to professors' own physics identities or sense of belonging during the literature search for this study.

E. Change theories

1. Professional learning communities

Contextualized research on how exactly a physics department should go about increasing belonging has been missing from the literature [15,20,49,50]. However, previous authors have stated that, to reduce participation gaps and improve outcomes for all students, physics instructors should engage in ongoing professional development about inclusive instructional strategies [4, 9, 20]. One way to deliver these types of change for professors is through professional learning communities [51], which offer collegial encouragement in addition to pedagogical strategies [52]. Such groups offer a venue for professors to examine their beliefs about teaching and learning, to consider how these beliefs affect and are affected by their interactions with students, and to support each other.

This paper describes part of a larger case study [17], focusing on the work that a small group in a certain physics department, which we here call The Committee, did to build community and increase student feelings of belonging. The research question for this paper is,

How did members of the physics department describe their experiences and perspectives on The Committee's work to increase student retention, diversity, and engagement?

2. First- vs second-order change

Substantial literature has been published in physics education calling for cultural change, without widespread success to date [19,31,42,43,45]. One explanation that has been offered is based on the concept of first-order versus second-order change [15,53].

First-order changes are surface level; they leave underlying social norms and belief systems unquestioned. An example is adding demonstration experiments to a class to illustrate concepts and generate interest. This can work well toward its intended goal, but it does not change the teacher-centered learning environment or require students to learn actively.

Second-order changes are much more challenging because they involve fundamental changes to the status quo. In the physics culture, believing that all students can succeed and that student-centered learning is valuable is a second-order change because elitism and lecture-only teaching are central to the traditional physics culture. Second-order changes require long-term commitment and expert support of the type that can be offered through ongoing professional development for faculty and staff [20,49] or via professional learning communities [15,50,54].

In this research, we examine the shift from the classical paradigm to the quantum paradigm in a physics department. This involves a shift in thinking about the types of changes that are required to improve belonging and retention in physics. Within the classical paradigm, first-order changes are sufficient to achieve these goals, while the quantum paradigm requires deeper, second-order changes that promote inclusive teaching and a supportive environment.

F. Qualitative case study methodology

This research is grounded in a primarily qualitative case study methodology for two reasons. First, taking steps to build student physics identities and feelings of belonging in most physics departments constitutes a significant cultural shift because it means moving away from traditional concepts such as meritocracy and lecture-only teaching, which are still prevalent [9, 16,38]. Cultural change, paradigm shifts, physics identity, and belonging are deeply human, context-dependent experiences that may be better understood using qualitative research methods. Second, research [15,20,54] has pointed to the importance of university departments, rather than whole institutions, as the key unit for improving student success. A university physics department, or a small group within such a department, is an appropriate choice for an in-depth, instrumental case study [55].

III. METHOD AND CONTEXT

Case study is the gold standard of research methodologies for investigating complex human phenomena such as belonging and cultural change [56]. This

research primarily focused on professors and staff in one Canadian physics department, although their students’ voices were carefully considered to create a deeper understanding of the case. The initial group of participants was the five members of an *ad hoc* committee that was formed with the goal of improving student engagement, diversity, and retention. This group included three professors, one staff instructor, and one undergraduate student, who we collectively call *The Committee*. The Committee was struck, in part, as a reaction to very low numbers of physics majors enrolled in their department’s undergraduate programs in late 2017.

A. Study context

The study took place at a medium-sized, research-based university in a department that offered several types of bachelor’s, master’s, and doctoral physics degree programs. The department was, based on the first author’s (L. S. C.) experience, similar to other physics departments in Canada. As is typical for this type of research, the degree to which readers feel that the results of this study apply to their own contexts will vary depending on the perceived similarities and differences between the study context and the reader’s.

During the study, L. S. C. was a doctoral student in education with 11 years of experience as a high school physics teacher and 7 years as a university physics instructor. She had been involved with qualitative physics education research for 5 years and had taught graduate courses in education for 2 years. She saw, and continues to see, herself as a member of both the physics culture and the education culture—two very different areas of academia. Much of this study involved work to bridge the gap between these two fields.

B. Data collection

Following ethical approvals, data collection took place over 17 months. The participant group was expanded beyond The Committee to capture a thick, rich description of the study physics department. More professors and instructors were recruited via networking and a teaching seminar series offered by L. S. C. All instructors in the department were invited to participate in the research, and the final instructor participant group approximated the instructor population of the department. The primary focus of the study became the 12 professor and staff participants, described in Table I .

TABLE I.
Description of the 12 professor and staff participants.

Rank	Tenure status	Course levels taught	Years taught
Sessional–3	Untenured, tenure track–3	First year only–6	None–1

Senior Instructor–2	Tenured–4	Upper year only–4	1 to 4 yr–3
Assistant professor–3	Tenure not available—5	Combination–2	9 to 12 yr–3
Associate professor–1			yr–5
Full professor–3			

The data collected from the professor participants included semistructured pre- and poststudy interviews, each of which lasted approximately 1 h. Further observations of the professor participants were collected during 11 one-hour teaching seminars, which they attended when possible, as well as individual meetings, meetings of The Committee, and a few class observations. A list of the teaching seminar topics is included in Appendix A . Detailed field notes were recorded throughout the research process, totaling 345 pages. Table II shows the timeline over which the various types of data were collected.

TABLE II.						
Timeline of data collection.						
	Date	Field notes: informal meetings, other interactions	Professor and staff interviews	Student surveys	Student interviews	PD seminars
2018	September	✓	✓			
	October	✓	✓	✓		✓
	November	✓	✓	✓		✓
	December	✓	✓			✓
	January	✓		✓		✓
	February	✓			✓	✓
	March	✓		✓	✓	✓
	April	✓			✓	✓
	May	✓				✓
2019	June	✓				✓
	July	✓				
	August	✓	✓			
	September	✓		✓		

October	✓		✓
November	✓	✓	✓
December	✓		✓

Student participants were a secondary focus of the research. Undergraduates at all levels were recruited from the classes of participating professors and invited to complete a voluntary, online, anonymous survey. This recruitment and survey process occurred 6 times in total—once at the beginning and end of each of the three semesters during which the research took place. Because the entire first-year teaching team and several key members of the upper-year teaching faculty participated, nearly every student in the department was invited to participate. Over the 6 iterations of the survey, 424 student responses were collected and included in the analysis. Due to ethical constraints, it was not possible to determine how many of those responses might have been from students repeating the survey at a different point in time; however, such responses are valuable because they represent potentially different perspectives.

The survey responses allowed the purposeful selection of student interviewees from a variety of demographic groups including males, females, a variety of self-identified ethnicities, physics majors, other majors, students from all years of study, and mature students. A total of 14 students participated in semistructured individual interviews that each lasted approximately 45 min. All student and faculty interviews were recorded and transcribed, and the transcripts were reviewed and approved by each interviewee to ensure their true words and meanings had been captured.

C. Data saturation

During the research, several of the professor and student participants inquired as to whether sufficient data were being gathered for the purposes of the study. When they learned that twelve professors and hundreds of students had participated, and that over 35 h of interviews had been conducted, they universally expressed that this was more than adequate to create a picture of their department. This perspective aligned with my own, since I (L. S. C.) had continued to recruit participants until I assessed that data saturation [57] was likely. The professor participants had between 0 and 25 years of teaching experience, some were tenured and permanent, and some were sessional. Between them, they taught almost every physics student in the university. The student participants had completed between 0 and 12 semesters of undergraduate courses. Most were in first-year physics courses, but many were in upper-year classes, approximately reflecting the population of undergraduate physics students. The population of student participants was slightly more heavily weighted toward upper-year students than the actual departmental population because the number of students in introductory classes was overwhelming, as it is in most Canadian physics departments.

D. Analysis

The data corpus was primarily qualitative and included over 1000 pages of transcripts and field notes. The analysis followed standards for qualitative analysis [58] and included the following phases [58]:

- 1) Familiarization with the data, involving reading, rereading, and noting ideas.
- 2) Initial (open) coding to determine what is being said in each piece of data and across the whole data corpus.
- 3) Axial coding. Searching for themes among the codes, relating and collating codes. This was done using code frequency counts, code meanings, and the comparative method [57].
- 4) Reviewing and relating themes, checking whether the themes accurately represent the data, creating an initial concept map.
- 5) Solidifying and defining themes, continuing to ensure the overall story is told accurately.
- 6) Reporting the results, choosing representative examples, relating to the literature and the research questions.

I (L. S. C.) performed the bulk of this iterative analysis; I collaborated with and was advised by K. M. in phases 4, 5, and 6. In addition, themes were recorded in the field notes as they arose during the research, which is recommended in qualitative research [49,59], especially during a project of this length. Throughout the analysis phase, codes and the data corpus were reexamined and revised as needed.

Every effort has been made to ensure the authenticity of the research as recommended by qualitative research scholars. Authenticity is the qualitative research parallel to validity in quantitative research. For example, this study's data were examined and compared across time periods, participant groups, and data types (qualitative and quantitative, transcripts and field notes, surveys and interviews) to achieve crystallization of the codes and themes [60]. This was done while including and carefully considering all data, including the dissenting voices rather than ignoring outliers [61]. Transferability has been accomplished to the extent possible via a thick, rich description of the study department [55] using maximum variation of participants across demographics, and sufficient data collection [62]. Reflexivity and an audit trail [62] provided further opportunities to ensure the integrity and reliability of the research.

The student surveys produced a modest amount of quantitative data, which were analyzed to determine whether and how they reinforced or refuted the qualitative data. Survey participation rates were low, at between five to ten percent of enrolled students each semester; however, they were high enough to obtain minimum confidence intervals of 90% and margins of error less than 10% for five out of the six surveys. These participation rates were not a significant

concern because the survey data were not the sole, nor even the main, method of data collection.

The student surveys contained ten questions that involved ten-point rating scales, as shown in Appendix B. The survey's aim was to help establish an overall picture of the student experience in the department; therefore, the questions were quite broad and were not validated. The quantitative results presented here were calculated using all student responses to find confidence intervals and p values [63]. A Bonferroni correction [64] was applied such that differences were considered significant for $p < 0.005$. Gender data are reported only for participants who self-identified as female or male due to very low numbers of participants who self-identified as other genders. The survey results were always compared to and cross-correlated with the qualitative data to provide authenticity to the study.

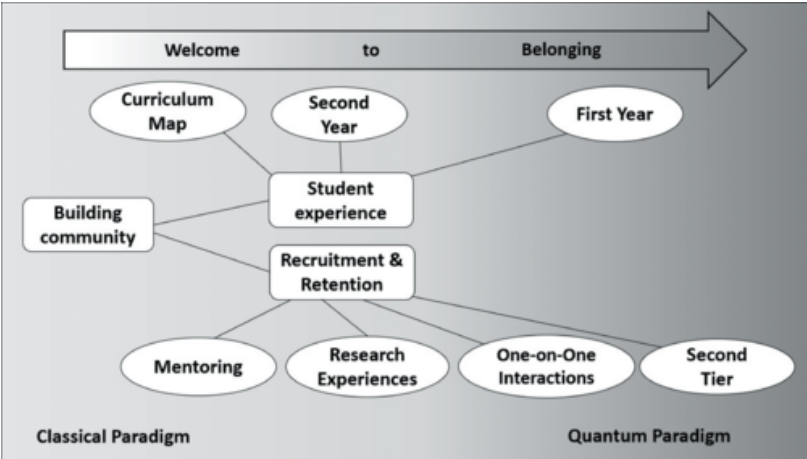
It is important to note that this study's data collection ended in December 2019, just prior to the COVID-19 pandemic, which had many unprecedented effects on higher education. With widespread returns to the classroom now having occurred, the results can be interpreted and applied in a postpandemic light.

IV. RESULTS

The research question asked how members of the physics department described their experiences and perspectives on The Committee's work to increase student retention, diversity, and engagement. One important theme in the data was the idea of building community, and this section describes this large theme and its smaller subthemes.

Figure 1 shows the relationships among the themes, each of which is presented in a bubble and positioned according to how the participants described it in relation to the other themes. Table III gives details about the themes in Fig. 1. Each theme is presented and described in detail in this section. This answers the research question by describing and relating the various aspects of the experiences that faculty and student participants reported.

FIG. 1.



Concept map showing the themes relating to building community that arose in the study data, and the steps faculty, staff, and student participants took in their efforts to increase student persistence and belonging. The light-to-dark gradient from left to right represents the paradigm shift from classical (traditional) to quantum (progressive).

TABLE III.

Descriptions of themes.

Theme	Description	Appeared in student data	Appeared in professor data
Building community	The need or desire to establish and maintain a group of peers for support and companionship during a difficult period, both for students during their physics degrees and for physics professors making changes in their work.	✓	✓
Student experience	The student’s view of the sum of everything that happens to them in undergraduate physics, both in class and outside of class. Faculty and student participants believed that improving student retention would mean improving the student experience.	✓	✓
	Broadening and increasing participation in physics by bringing		

Recruitment and retention	more students in (i.e., recruitment) and convincing more of them to stay (i.e., retention). This was the Committee's original purpose, but professor participants found it more complex than they may have expected.	✓	✓
Welcome to belonging	Changing perspective from allowing anyone to attempt to succeed in physics in its current form to encouraging everyone to stay in physics because they are valued. Happened approximately in parallel with changing from classical to quantum paradigms.	✓	✓
Curriculum map	A how-to guide that some departments develop to show undergraduates various ways of obtaining a degree through taking different groups of courses in a certain order. There was a widespread agreement among faculty and student participants that making one was a good idea and would help students stay in physics and have a better experience.	✓	✓
Second year	The two semesters of a physics degree program that follow the completion of the full first year of introductory physics. Most faculty and student participants believed that second year students were part of the physics department, and that second year students belonged to some degree.	✓	✓
First year	The first two semesters of undergraduate physics, often taken immediately when students enter university. Most professor and student participants did not believe that first year students belonged to the physics department, at least not yet.	✓	✓

Mentoring	A professional relationship between a student and a professor or staff member that is informal in nature and provides the student with encouragement and guidance. A previously successful mentoring program in the department was expanded during the research, and there was broad agreement that it was valuable.	✓	✓
Research experiences	Student work terms, often completed during the summer, wherein they participate in faculty research projects and learn how to conduct academic studies. Student and professor participants named these as highly influential for students who had them.	✓	✓
One-on-one interactions	Any verbal or nonverbal communication between members of the physics community, sometimes between a professor and a student, either during class or outside of it. Some of these exchanges were noted as memorable or important by student and professor participants, even if they felt the other party involved would not remember it.	✓	✓
Second tier	Students who are capable and interested in physics but may be overlooked by their professors if their grades are less than top-notch [42]. Some student and faculty participants implicitly or explicitly stated that these students did not belong in physics.	✓	✓

Note that the background of Fig. 1 changes from light gray on the left to dark gray on the right, which represents the shift from the classical paradigm on the left side of the figure, to the quantum paradigm on the right.

A. Building community

The concept of building community appeared many times in the study data. It was important to both student and professor participants, and they described their experiences from both positive and negative perspectives.

Student participants described how having a supportive community that included their peers and their professors helped them to succeed, or conversely, how the lack of such a community was detrimental to their experiences in physics. One female student in first-year engineering reported that, "I couldn't imagine trying to do physics on my own. ... I always feel supported. ... We're always there for each other. [The Engineering department members] really encourage the teamwork." A converse example came from a female, potential physics student in the same class, who noted,

I did not make any sort of peer relationships. ... I think it would have been such a different environment had [someone said], hey, you guys [physics majors] are here devoted to learning about this ... That might have actually kept me [in physics].

The difference between the engineering student and the physics student's experiences starkly demonstrates the difference a feeling of belonging can make, especially if encouraged by faculty and program administrators. Again, these two students were members of the same class, in the same year, and were both women.

In addition to students' comments about belonging, several professor and staff participants stated that they often felt isolated. In a prestudy interview, one professor said, "It's not a strong teaching team, a sense of teaching as a team. ... There's a need for conversations between profs about what we do and how we do it, and what's the best strategy." The few professors who had previously participated in professional development about teaching unanimously stated how beneficial it had been for them because they had been part of a professional learning community (PLC). A PLC was established in the department as part of this research, and by the end of the study, all those who had participated in it, and several who had not, espoused the value of being supported by their colleagues. One professor stated that participating in the study had created "an opportunity to do a lot of thinking about teaching, and to discuss different ideas with you [L. S. C.], and with my other colleagues as well. And I have found that immensely useful." Because they either valued being part of a supportive community, or because they felt they lacked one, student and professor participants confirmed that feeling that they *belonged to a community* in physics was highly valuable.

In addition to this desire for supportive communities among students and faculty, there was interest in creating a stronger department-wide community, including undergraduates, graduate students, staff, and faculty. A professor participant stated at the beginning of the study that, "We need to be [a] community with [the students], and that we don't have." At the end of the study, this same professor noted improvement in this area, saying,

I think it's a matter of being more present with each other as a team, and being more present with the students. ... The couple of social events that have been

done so far, I think have been very well received.

When the research concluded, The Committee had a plan to host eight annual community events in the year following the study, because their initial efforts with such events had increased the feelings of community that many participants felt.

During the data analysis, the *building community* theme was resolved into three parts: moving from welcome to belonging, improving the student experience, and increasing recruitment and retention. These are shown in Fig. 1 and described more fully in the following sections.

B. Welcome to belonging

A female student interviewee said,

It's very welcoming here, I think. But it could just be that I fit in. Maybe someone else wouldn't feel as though they fit in. There is definitely a sense that you've got to prove yourself before you're taken seriously, which is unfortunate. ... Maybe that's a female experience.

This comment describes this student's perspective of the difference between welcome and belonging: she felt that everyone was welcome, but that belonging required a demonstration of competence.

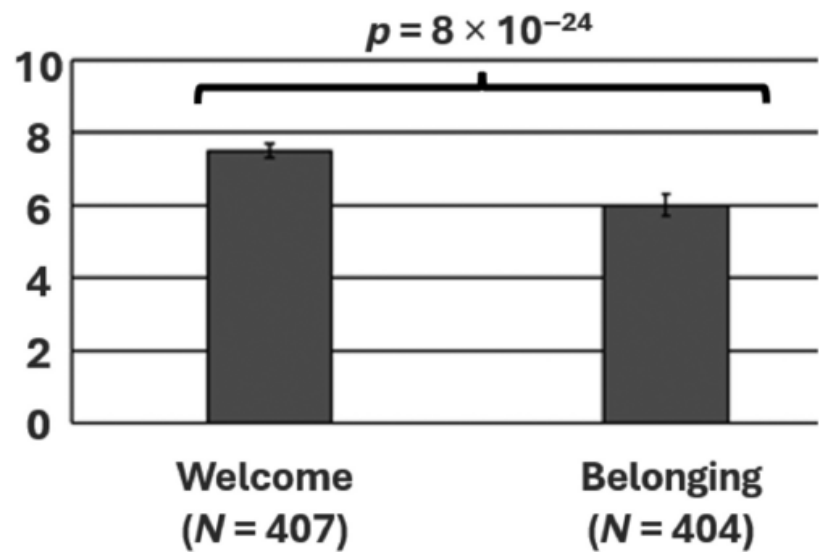
1. Differentiating welcome vs belonging

Our results suggest that there is a difference between feeling welcome in a physics department and feeling that one belongs in a physics department. This idea is new and arose based on both qualitative and quantitative data we collected. This barrier between welcome and belonging has not been previously shown in physics education research, although it was mentioned in a report [43], it confirms a conclusion of [33], and it has appeared in literature not specific to science, technology, engineering, and mathematics [2].

During the data collection, we did not define the terms "welcome" or "belonging" for participants. This means that they interpreted these terms according to their own experiences and/or the commonly understood definitions described in the theoretical framework above.

In the survey data, student participants reported feeling quite welcome in the department (out of 10); this should be celebrated. However, as a group, students felt a significantly lower sense of belonging (), as shown in Fig. 2 .

FIG 2.



Student survey response to survey questions 1 and 2 in Table IV, ranked between 1 and 10. Error bars indicate the 95% confidence intervals.

In addition to this difference for all students, the survey results showed significantly lower responses from female students compared to their male counterparts on eight of the ten questions, as shown in Table IV. This means that female students did not feel that they fit as well into the department as male students felt, and therefore, they were less likely to persist in physics. This confirms much prior research [10,12,19,21,23,30,31].

TABLE IV.

Statistically significant differences in male vs female student survey respondents about their experiences in physics.

	Question	Male student respondents	Female student respondents	value
1	How welcome do you feel in the physics department?			
2	To what degree do you feel you belong in the physics department?			
3	Overall, how comfortable do you feel with your classmates in your current physics course(s)?			
	Overall, how comfortable do you feel in your relationships			

4 with professors that you know
in the physics department?

Overall, how comfortable do
5 you feel in your relationships
with other physics department
staff that you know?

To what degree do you feel that
8 your current physics classmates
would like you to continue in
physics after this course?

To what degree do you feel that
9 your current physics
professor(s) would like you to
continue in physics after this
course?

10 How likely are you to continue
in physics after this course?

These quantitative results would have been difficult to interpret alone; however, the qualitative data provide a much richer picture.

Previous research has shown that professors can have a profound positive influence on students' physics identities by explicitly recognizing them as members of the physics community [3, 4, 9]. However, in interviews for this study, many of the professor participants expressed concerns about whether it was appropriate, or even possible, for them to encourage students to feel belonging in the department. Three of the professors felt that their actions did not affect student persistence in physics in any way. Two faculty participants had concerns about how and whether they could ethically build relationships with students, while three thought it inappropriate to encourage students they considered less than top-notch. The idea of encouraging and supporting students was challenging for many of the faculty participants. This concern was echoed by several upper-year student interviewees who questioned whether students who needed encouragement could really succeed in physics, anyway. These strongly held beliefs in the physics culture illustrate the difficulty of making the paradigm shift to inclusion and belonging.

Another perspective on this issue came from the many student and professor participants who reported that they did not believe that they personally belonged in this physics department. Several made statements to the effect that they were not "real" physicists. Of the 12 professor interviewees, 6, including both of the female professor participants, reported that they did not feel they belonged in physics. For example, one professor said,

I'm not really a physicist, [although] I did take a number of physics classes in my undergraduate degree. ... I feel like a second-class citizen in this department. ... I was a Chair at [an oral] exam, and somebody actually said, pretend you're explaining this to [speaker's name], who doesn't know anything. And I thought, you know, ... it's been [a few] years, but I do know about this stuff. So I've never chaired another one of those exams again.

This professor had been deeply affected by the dismissive words of a colleague based on a minor difference in training. Of the five other instructors who reported feeling the same lack of belonging, for two, this was due to similar small differences in training, and for the three others, it was because they were in nontenured positions. This is a new result; the lack of belonging among professors has not been noted in the physics education literature prior to this writing, though it has been alluded to in research with women physicists from underrepresented groups [32,50].

Rooted in the traditional physics culture, this sense of not qualifying as a real physicist, and indeed of gatekeeping ourselves out of the physics community, reflects elitism, fragile physics identities, and a lack of community support. Participating in the research, and specifically in the professional learning community we developed, helped the professor participants to feel more confident in their teaching and built a greater sense of belonging in the department. Several participants voiced a strong interest in continuing to meet with this group after the research concluded.

The white-to-gray gradient of the welcome to belonging arrow in Fig. 1 reflects that welcoming students to the department, shown on the left side of the figure in white, was a comparatively easy concept for the study participants to accept and implement without changing their classical paradigms. Indeed, the student survey results indicated that, to a reasonable degree, this had already been achieved. Fostering belonging, on the right side of Fig. 1, in gray, proved quite a bit more difficult for most participants and seemed to require a larger, cultural shift to the quantum paradigm.

This research reveals a gap between participants' feelings of welcome and belonging in physics. The latter were more difficult to develop, and doing so may require a paradigm shift. Furthermore, for professors who do not, themselves, feel that they belong in physics, we can expect that helping students feel a sense of belonging will be even more complex.

C. Student experience

Many participants, both students and faculty/staff, discussed their desire to improve student experiences in the department. One professor said simply, "We don't have enough students. So... we need to give students good experiences." Another stated that, moving forward after participating in the research, his goal

was to remember that his students “have to know their stuff, and they have to enjoy the process at the same time.” Several aspects of improving the student experience arose as themes in the qualitative data. These aspects, shown in Fig. 2, and described further here, were a curriculum map, the second-year student experience, and the first-year student experience.

1. Curriculum map

Many student and professor participants felt that creating a coherent curriculum map for the undergraduate program would improve the student experience. If a department has considered the various possible pathways through its degree program, and explicitly delineated these for students, this can give students confidence that they are completing the required elements of their programs. The process of developing such a document had begun in the department before the research began and continued after the study concluded, demonstrating that it was, as one professor called it, “a difficult optimization problem.” Because participants unanimously agreed that a curriculum map was important to the student experience, it is located on the far left of Fig. 1, within the classical paradigm and among the more traditional and easily accepted ideas.

2. Second year

Improving the experiences of second-year students was similarly viewed by most participants as important because second-year students are poised to continue into physics for the longer term. One professor said that in the department, I think everybody agrees that we need to work on our second year. Trying to find a way to make it, not easier, but more appealing somehow. [We need to] make sure that they [students] get what they need, they get some physics out of it, and it's a good experience.

To this end, changes were made to the second-year program during the study with the goal of encouraging persistence. For example, the second-year lab course was redesigned to provide greater opportunities for authentic experimentation and active learning. In addition, three professors stated that they wanted to try to ease students' transitions between first and second year. Second-year students and upper-year students were regularly included in departmental events, indicating that they were valued as members of the community.

Most of the professor and student participants stated that they believed second-year students were part of the physics community and therefore deserved the department's support. Offering this support by making improvements to the second-year program, therefore, still fit within the classical paradigm. However, these changes were a little more challenging for some participants to accept than the curriculum map because they did involve some movement away from traditional teaching and toward student support. For this reason, second year

appears in Fig. 1 near the left side, but to the right of the curriculum map, as it is transitioning toward a quantum paradigm.

3. First year

Here, we begin to encounter more significant disagreement among participants as to who does and does not belong in the physics community. Specifically, both professor and student participants were split about whether first-year students were truly members of the physics department. One longtime first year professor stated that in the large first year classes,

A lot of those are pretty weak students and they definitely would not succeed in a physics program. You know, 75% of the class is engineering... The majority of the class, they're not physics people.

An upper-year physics major agreed, saying,

I really don't feel like first year physics students are part of the physics department ... A lot of people haven't even declared their majors in first year. They're in one physics course; it's not like they're really a part of the department until second year.

These points are worth considering, and they are common among many members of the physics community. They reflect the traditional physics culture and the classical paradigm.

But not all participants agreed. Some, including all of the first-year students who had considered majoring in physics, and a few instructors, believed that first-year students should be included in the department. One professor pointed out that, "The biggest retention issue is from first year to second year, and there we're not doing anything because we don't even know that the students are interested in physics until they show up in second year."

Paradigms began to shift slightly in this area during the study. For example, an existing mentoring program was expanded to include interested first-year students. "As a department, we are extending out to first-year students, and inviting them [in]," one professor noted proudly. However, at the end of the research, the idea of including first-year students in the physics community was not widely accepted by the participants. The study data showed that the belief that first-year students do not yet belong in physics was strongly held, and changing it required a paradigm shift. Therefore, we categorize it here as part of the quantum paradigm.

D. Recruitment and retention

There was general acknowledgment among participants that more students were needed in upper-year physics courses for the department to sustain itself. In addition to their concerns about numbers of students, many professors

worried that the recruitment and retention problem would not be easily solved with familiar strategies such as outreach programs. For example, one professor said, “What we could have done 10 years ago, and these are [still] some of the things we’re thinking about today, are not enough now. It’s already—We need to be far more aggressive than that.”

In the qualitative data, participants described their concerns about recruitment and retention, and four major subthemes arose. These are explained below and shown in Fig. 1.

1. Mentoring

A departmental initiative that many student and faculty participants recognized as successful was their small group mentoring program. This extracurricular, informal program had been open to second- and upper-year students before the study began, but it was expanded to include interested first-year students during the study. Student and professor participants reported finding the mentorship program valuable and time efficient. One professor said,

Compared to other methods, mentorship groups are so personal... The effective time spent is really high. ... I think that just having those small groups, where the students are just so personally connected to [a faculty member], is really effective.

Small group mentoring offered the chance to build supportive relationships outside of classes, while fitting well into existing professor and student paradigms. Therefore, this theme is placed on the left side of Fig. 1 and is part of the classical paradigm.

2. Research experiences

Professor and student participants broadly agreed that undergraduate research experiences helped students build their sense of belonging in the physics community. Such experiences are part of the traditional physics culture for high-achieving, upper-year students, and participants universally acknowledged the benefits of undergraduates participating in research. However, the prospect of including first-year students in these opportunities was difficult for most students and professors to accept. Other concerns around undergraduate research included the financial resources and time required from faculty to implement it, and the possibility that low-quality research would be produced. The consensus was that research experiences were incredibly valuable, but offering them to a wider variety or a larger number of students was problematic. Despite these issues, we include undergraduate research experiences as part of the classical paradigm because they align with the traditional physics culture for successful upper-year students.

3. One-on-one interactions

The study data included many instances when an interaction between a student and an instructor or teaching assistant had important and lasting effects on the student's physics identity or sense of belonging. One student who reluctantly left the physics major explained part of her perspective, which she had developed over several semesters.

In chemistry ... it definitely felt like I was more of a person or an individual than I did in physics. I think that might be it—it felt like [the chemistry professor] cared and that I wasn't just another student she had to deal with.

This experience contrasted with times where this student felt ignored and marginalized by her physics professor. Unfortunately, some of the physics faculty, including the above student's professor, incorrectly believed that they had little to no influence on student recruitment or retention. When asked about what they were doing to improve student recruitment and retention, a first-year instructor said,

For me, in first year, to attract and retain students, we don't really have too much control over attracting. That comes from enrollment in the university. Retaining, you know, our midterm averages are around 55 percent. ... A lot of those are pretty weak students and they definitely would not succeed in a physics program. ... I think we get enough in the second year that would more than satisfy the needs of the upper level program, if they were retained.

In hopes of addressing this "enrollment in the university," a long-standing outreach program was in place in the department. However, during their interviews, a few professors and staff questioned the program's effectiveness, as there were no methods to track people's reactions to the outreach or the reasons why students enrolled in physics.

Some upper-year professors shared the belief that they did not affect student recruitment or retention, with one saying, "Normally I never worry about retention or anything like that. They're basically set at this point ... By the time I see the students, they're very close to finishing."

However, as part of the study, a student who had just finished third year was interviewed. This student expressed several complaints about the department but seemed overall satisfied with the program. Unfortunately, in an email a few weeks after the interview, the student wrote, "I no longer have any desire to stay in Physics. I intend to take [a few weeks] and assess whether or not I wish to continue. I do not think I will." This student had outwardly seemed "basically set," but their departure from physics after 3 years demonstrated that students, at all levels, require support, contrary to what the professor participants believed.

Adding to this, numerous student participants remarked on how they were directly influenced by their instructors and teaching assistants to persist in or to leave physics. During interviews, three out of the five undergraduate physics

majority reported that an important reason why they were pursuing physics degrees was because one or more of their professors had explicitly encouraged them, either in person or by email. Some of the first-year instructors reported having made past efforts to directly recruit students into physics degrees, with varying levels of success. Other professors reported that they were not comfortable having such conversations.

There was considerable disagreement in the data about whether professors are responsible for recruitment and retention, and whether and how they should professionally approach relationships with students. Therefore, building skills around one-on-one professor-student interactions is placed on the left side of Fig. 1 and is part of the quantum paradigm.

4. Second tier

In both the quantitative and qualitative data, a second tier of students appeared and confirmed the existence of such a group as was discussed in 1990 by Tobias [42]. This study's second tier appeared in the student survey data when those with self-reported grades in the B range reported feeling significantly less welcome () and less belonging () than those who reported earning A grades. More positively, second-tier student participants reported similar levels of interest in continuing in physics () as compared to their top-tier counterparts, which further confirmed Tobias' assessment that there are interested and capable students who are not being encouraged under the classical paradigm.

This quantitative result was confirmed several times in the qualitative data. For example, student interviewees were asked, "Is physics only for the best and the brightest?" and their responses were telling. Of the 14 students interviewed, 3 said yes, this was true. Eight more gave a "no, but" response, which was followed by a caveat such as, "you have to work very hard," "you have to be passionate about it," or "you have to be able to think a certain way."

Many of the professor participants voiced similar views, although they were not asked the specific question about the "best and brightest." One example is that a first-year professor asserted that using research-based instructional strategies aimed at the majority of students "doesn't change things for the top end. ... They're the ones we love, right, and the ones who go on [in physics]." Another professor agreed, saying, "These people aren't going to be retained in physics, the majority of them ... no matter what we do." These responses, when considered together, demonstrate a pervasive, normally unspoken view that physics is indeed for the elite of academia.

However, not every professor participant was willing to disregard the second tier. With support from the first author, two professors made substantial pedagogical changes during the research which aimed to benefit more students with midlevel grades. One implemented a new lab program that emphasized group

work and mastery of concepts rather than lab report writing, and the other adopted a flipped classroom approach that centered on active student learning instead of lecturing. Both of these instructors later stated that the changes they made had been valuable for the majority of students and for themselves as instructors. Both acknowledged that they had spent a long time thinking about implementing such changes but would not have moved forward without support. The paradigm shift into progressive teaching was daunting, but those who undertook it reported important benefits, both for themselves and for their students.

The study data show that providing the encouragement, support, and community required to retain second-tier students was a concept with which most faculty and student participants struggled. This difficulty places it in the quantum paradigm, yet the experiences of the professor participants who attempted to encourage and support second-tier students demonstrated the profound benefits of this strategy.

V. DISCUSSION AND IMPLICATIONS

The physics department in this case study had many positive features, including an active interest in improving the retention, diversity, and engagement of students in their classes, a successful mentoring program, and a high rating from students in feelings of being welcome. If readers do not believe that their own departments already possess these attributes, this research suggests beginning with developing them, as the rest of this study’s results built on the extant culture of the department. Creating conditions on the left side of Fig. 1 lays the foundation for the development of the ideas further to the right and increases the likelihood of sustainable change. Participants’ journeys did not always move linearly from left to right on the figure, but establishing or strengthening the classical approaches described in the Results section and listed in Table V is important for readers hoping to begin or continue moving into the quantum paradigm, building community and addressing student retention.

TABLE V.

Strategies used by study participants within classical and quantum paradigms.

Strategies within the classical paradigm	Strategies requiring a quantum paradigm
Curriculum map	First year inclusion
Second year program improvements	Improved one-on-one interactions
Mentoring	Second tier inclusion
Undergraduate research experiences	

Welcome

Belonging

The study data demonstrated a new difference between students' feelings of welcome and of belonging, with belonging being more difficult to achieve. This lack of belonging was worse for female students compared to their male colleagues. We recommend that interested members of the physics community consider how they can help all students, but especially women and other underrepresented students, not only to feel welcome but also to feel that they belong in physics. This means providing a caring community that believes each student has something of value to contribute to physics.

Importantly, the sense of not belonging, and its gender divide, extended to professor and staff participants in this study. The shared, yet isolating, feeling of not being a real physicist, among people who would certainly be considered by the public to be physicists, demonstrates the fragility of physics identities and the insidiousness of the elitism in physics. The tacit belief that one must follow a specific educational and career path, or that one must consistently demonstrate brilliance without support, to fully belong in the physics community, is a myth based on a traditional, elitist physics culture that many continue to deny exists [26,35].

If the sense of belonging in physics is so difficult to achieve, does this explain why so few people choose to pursue physics degrees? Until we, as a physics community, can offer an environment in which all interested people can feel that they belong, it is disingenuous and harmful to continue to ask why retention is so low, and why there are not more women or members of other underrepresented groups in physics.

Belonging is a critical human need [65], important for the development of strong physics identities [1, 21], which in turn leads to greater persistence in physics. This is especially true for women, racialized people, and members of other underrepresented groups [1, 28, 30, 33]. Therefore, improving students' and professors' feelings of belonging in physics may be critical to diversifying and sustaining the discipline. This study confirms, however, that improving belonging in physics is complex and difficult. It requires a fundamental, quantum shift in how members of the physics community think and act.

The participants in this research found significant success in building community on several levels: among students, among faculty and staff, and across the whole department. This confirms prior research [6–8, 43]. Community building was relatively easy to implement in this study, at least as far as classical paradigms would allow, and it should be a priority for any physics department wishing to increase, maintain, or diversify enrollments. Participants were generally enthusiastic about encouraging and supporting second- and upper-year students, and about supporting each other.

Including first-year students in the physics community was more controversial, but neglecting the needs of first-year students has been shown to be detrimental not only to those students but also to departments [66–68]. This study and others [3, 42, 69] demonstrate that many students succeed in first-year physics with grades just short of excellent, and they would be interested in pursuing physics degree programs if they were minimally supported. Encouraging such students, perhaps through one-on-one conversations, with messages from faculty, or via community events, is an important area for future growth. These ideas can be implemented with a nominal investment of time and resources and may help students consider, perhaps for the first time, that they could belong in physics and do a physics degree.

The idea of supporting and encouraging students was, however, difficult for some professor participants to accept. Their reluctance was based on one or more of the following: (i) the belief that those who are smart enough to succeed in physics will do so without support, (ii) a gap in their understanding of how to appropriately offer that support, or (iii) a belief that professor's actions did not affect student persistence at all. These ideas are part of a traditional, classical physics teaching paradigm. The literature is clear that professors profoundly affect student experiences in physics, either positively or negatively, whether they intend to or not [9, 12, 19]. Disregarding or minimizing this effect is an example of the traditional physics culture creating an unnecessary barrier to belonging, sustaining elitism in our field. Study participants who did attempt to encourage and support students had some understanding of the quantum paradigm and were generally more open to second-order changes. One way that participants' traditional beliefs were addressed in this study was through professional development in a professional learning community, which additionally provided a source of support and belonging for the professors themselves.

This research suggests a reason why increasing student enrollment in physics is difficult and complex: There is a limit to improving student recruitment and retention if classical methods such as those listed in Table V are the only approaches used. Just as in classical versus quantum physics, some problems are better tackled using a quantum paradigm, which here means a fundamental change toward more progressive, inclusive ideas of education. This includes acknowledging that professors' actions and beliefs critically influence student experiences, both positively and negatively. Coming to this understanding is a second-order, quantum change that requires members of the physics community to accept that our elitist, classical culture is flawed; it creates barriers that prevent more people from pursuing physics. The classical physics culture drives away the same people that we claim to want to attract and keep in physics. Acknowledging this problem and accepting responsibility for changing it are the first, difficult steps in the journey to resolving it and making room for everyone who wants to participate in physics.

VI. LIMITATIONS

This paper represents a part of a larger study; for greater detail and context, please see Ref. [17]. In particular, Ref. [17] includes a detailed discussion of how participants variously engaged and struggled with shifting their mindsets toward more progressive understandings of education, and a full analysis of their perceptions of traditional versus progressive classroom teaching methods.

The research was primarily qualitative in nature, which means the results may not directly transfer into different contexts. The study was conducted according to best practices in qualitative research [62], including close attention to authenticity, crystallization from multiple data sources, and an audit trail.

Participation in the research was sufficient to make the assertions in this paper. Within the bounds of the research ethics permissions obtained for the study, the number of participants was maximized. Greater participation, however, may have deepened or broadened the results. Similarly, a longer time frame for data collection may have expanded and strengthened the study's conclusions.

The quantitative data from student surveys that we have presented should be interpreted with caution for several reasons. For example, the survey questions used have not been validated. However, the wording of the questions was similar to parts of the validated survey used in Ref. [3]. Further, no inferences were made in this study using solely quantitative results; these were used only in combination with the substantial qualitative data.

The data for this study were collected from September 2018 to December 2019, just prior to the global COVID-19 pandemic. Many aspects of university education and academic communities have changed since that time. However, the messages of this research remain relevant postpandemic and may be even more important for racialized students [70–72]. As we now work to rebuild our departmental communities and cultures, we have an opportunity to include more people in physics.

VII. CONCLUSION

This study's participants made progress toward building community among students, among professors, and across the whole department as a means of increasing their enrollments and diversity. Improving the student experience was one aspect of this progress, including efforts to create a coherent curriculum map, making improvements to the second-year program, and increasing access to the department for first-year students. Another part of the community building efforts focused on student recruitment and retention, including a mentorship program, undergraduate research experiences, one-on-one interactions, and supporting the so-called second tier of students. Participating in community building helped some participants move from welcoming people into physics to encouraging greater feelings of belonging. This type of work aims to strengthen and diversify membership in the physics community for the long term, but this research confirmed prior work [14,15,49,50] in that the changes

were difficult, took a long time, and required support.

Figure 1 is perhaps the most important result in this paper. Readers are encouraged to consider where, given their own situations, some effort might be best spent to make progress toward the right side of the figure. Considering the literature on first-order versus second-order change, we suggest that interested parties first consider building or strengthening the “classical” approaches to building community listed in Table V. With such a foundation, paired with interest and motivation, the “quantum” paradigm may then have greater likelihood of success. Study participants helped build a strong community to increase student recruitment and retention by engaging in incremental shifts of their departmental culture toward diversity and inclusion.

Positive changes were possible while maintaining a classical paradigm about physics teaching and culture. However, the classical paradigm limited what participants were willing to consider in terms of change. Beginning to shift into the quantum paradigm was necessary for the second-order changes that some participants made and that other authors have indicated are important developments for the physics culture [3, 9, 12, 27]. The challenge of making this paradigm shift is an important aspect of the complex barriers to participation that the traditional physics culture creates. Even if members of the physics culture accept the need for change, making the shift toward belonging and a community of support may be daunting, especially for those who do not already feel a sense of belonging themselves. We suggest that readers interested in making the types of changes in this research should seek out a community of support for themselves among colleagues in physics, science, education, and other university departments.

This study revealed a new gap between participant feelings of welcome and belonging in physics for students, and a similar lack of belonging among professors. Understanding this gap, and the associated challenge of shifting paradigms in the physics culture from classical to quantum, will help increase recruitment, retention, persistence, and diversity and will move the physics culture toward wider support and inclusion for all.

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The authors are profoundly grateful to the study's professor and student participants, and the physics department where it was conducted, without whom this research would have been impossible.

Appendices

APPENDIX A: LIST OF TEACHING SEMINAR TOPICS

These were the topics of seminars that were offered monthly during the research. The sessions allowed professors in the department to meet, learn, and support one another in a professional learning community. The topics were

chosen by the first author, sometimes based on suggestions of the participants.

1. Course planning and your syllabus.
2. How to write course outcomes.
3. Are you with me? Soliciting student feedback with high-tech, low-tech, and no-tech approaches.
4. Encouraging students by being yourself: How your passion for science can make a big difference for students.
5. The physicist's apprentice: The incredible value of authentic undergraduate research experiences.
6. Do they get it? Formative assessment as a tool for both professors and students.
7. Back to Basics: Lesson planning and best practices for presentations.
8. Writing, Teaching with, and Assessing Conceptual Physics Questions (or, Agh! Where are the numbers??).
9. Avoiding Blank Pages: Teaching students how to start an unfamiliar problem.
10. Your teaching crew: Who's in it, and why it's important to have one.
11. Revitalizing Your Teaching: Easier and more beneficial than you might think.

APPENDIX B: STUDENT SURVEY QUESTIONS

1. Section 1: Demographics

1. Your year of study:

1st, 2nd, 3rd, 4th, 5th, 6th, other (please elaborate) [open text response]

2. To which gender identity do you most identify?

Female, Male, Transgender Female, Transgender Male, Gender variant or non-conforming, Other (please elaborate) [open text response].

3. Your self-identified visible minority status (based on Statistics Canada categories):

Aboriginal, South Asian, Chinese, Black, Filipino, Latin American, Arab, Southeast Asian, West Asian, Korean, Japanese, Multiple Visible Minorities, Not a visible minority, Other (please elaborate) [open text response].

4. Are you a physics or atmospheric science major?

Yes No Not sure (please explain).

5. How many semesters of university level study have you completed to date?

1, 2, 3, 4, 5, 6, 7, 8, 9, other [open text response].

6. What grade do you anticipate receiving in your current physics course(s)?

_____, A, _____, B, _____, C, _____, D, F, Don't know.

2. Section 2: Your views

These questions ask you to consider your current thoughts about yourself, physics, and the ***[redacted]*** physics department. Please answer according to your current perspective, today, even though your thoughts may have been different in the past.

On a scale from 1 to 10, with 1 meaning “not at all” and 10 meaning “extremely,”

1. How welcome do you feel in the physics department?
- 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.
2. To what degree do you feel you belong in the physics department?
- 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.
3. Overall, how comfortable do you feel with your classmates in your current physics course(s)?
- 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.
4. Overall, how comfortable do you feel in your relationships with professors that you know in the physics department?
- 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.
5. Overall, how comfortable do you feel in your relationships with other physics department staff that you know (e.g., lab instructors, lab technicians, teaching assistants, administrative staff)?
- 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.
6. How successful do you feel that you have been in your most recent physics course(s)?
- 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.
7. Up until today, to what degree do you feel that you have lived up to your potential to succeed in physics?
- 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.
8. To what degree do you feel that your current physics classmates would like you to continue in physics after this course?
- 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.

Outline	Information
Citing Articles (1)	
Abstract	
Article Text	
INTRODUCTION	
THEORETICAL FRAMEWORK	
METHOD AND CONTEXT	
RESULTS	
DISCUSSION AND IMPLICATIONS	
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CONCLUSION	
ACKNOWLEDGMENTS	
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9. To what degree do you feel that your current physics professor(s) would like you to continue in physics after this course?

1, 2, 3, 4, 5, 6, 7, 8, 9, 10.

10. How likely are you to continue in physics after this course?

1, 2, 3, 4, 5, 6, 7, 8, 9, 10.

11. Compared to last semester, how do you feel about continuing in physics?

Better, Worse, The same.

If better or worse, please explain why. [open text response].

12. Will you allow the principal investigator to contact you regarding participating in an interview about your experiences in physics?

Yes, No, Other (please explain) [open text response].

If yes, please enter your email address here: [open text response].

13. If you would like to receive a copy of the final report for this thesis or any follow up publications, please send a separate email to lstyles@stfx.ca.

14. Please feel free to add any additional comments you may have here. [open text response].

APPENDIX C: STUDENT INTERVIEW GUIDING QUESTIONS

1. Tell me a bit about your education to date. For example, how long have you been at **[university name redacted]**?

2. What are your thoughts about where you are going in your education and career? (Do you think you will continue in a physics degree program? Why or why not?).

3. Have you had any memorable people or events during your physics education that really affected you?

4. Why did you complete the questionnaire and offer to participate in an interview?

5. What would a perfect physics course be like?

6. If you could change anything about this department, what would you change and why?

7. How do you think more students could be encouraged to continue in physics degree programs?

8. Do you feel welcome in the physics department? Why or why not?

9. Do you feel supported in your chosen degree program? Why or why not?

10. How successful do you feel you have been in physics? Please explain.
11. Some people think that physics is only for the “best and brightest.” What would you say to them?
12. What are your thoughts about diversity in physics?
13. Is there anything else you’d like to mention or discuss?

APPENDIX D: PROFESSOR INTERVIEW QUESTIONS

1. Prestudy interview guide

1. Tell me a bit about your career to date. For example, how long have you been at ***[university name redacted]***?
2. How important would you say teaching is to you in your job?
3. What is a perfect day in your classroom like for you, and for your students?
4. What is your teaching philosophy?
 - a. How were you taught? What have you learned about teaching and how/when/where?
 - b. How important would you say teaching is as a part of your job?
5. If you could change anything about this department, what would you change and why?
6. What are your thoughts about student retention in physics?
 - a. Why did you join the Committee?
7. What are you doing now to attract/retain students?
8. What are your thoughts about physics education research?
9. What are your thoughts about student engagement and success?
10. What are your thoughts about diversity in physics?
11. If a student came to you and said she or he was considering changing from an honors physics program to a physics major, what would you say?
 - a. What if the student said she or he was leaving physics for another major entirely? Has this ever happened to you?
12. What can I do to help you?
 - a. What roadblocks do you encounter when you are teaching?
13. Is there anything else you’d like to mention or discuss?

2. Poststudy interview guide

1. What were the challenges and (or) benefits you encountered during the study?
2. Looking back over the duration of this study, has your perspective changed,

and if so, how?

3. What are you doing now to attract/retain students?

4. Would you say that students are more engaged in your classes now than at the beginning of the study? Would you say that their grades are better? Would you say that the retention rate has improved?

5. How important would you say teaching is to you in your job?

6. What is your teaching philosophy?

7. What is a perfect day in your classroom like for you, and for your students?

8. If you could change anything in the department, what would you change?

9. What are your thoughts about student retention in physics?

10. What are your thoughts about student engagement and success?

11. What are your thoughts about diversity in physics?

12. What are your thoughts about physics education research?

13. If a student came to you and said she or he was considering changing from an honors physics program to a physics major, what would you say?

a. What if the student said she or he was leaving physics for another major entirely? Has this ever happened to you?

14. What would you share with your physics colleagues about this study?

15. What's your plan moving forward?

16. Is there anything else you'd like to mention or discuss?

APPENDIX E: DEMOGRAPHICS OF STUDENT SURVEY RESPONSES

Gender ^a	Year of study ^a	Ethnicity ^a	Physics major or not ^{a, b}	Estimated physics grade(s) ^a
Female – 48.1%	1st – 48.0%	Aboriginal – 1.7%	Yes – 18.9%	– 16.2%
Male – 50.2%	2nd – 22.3%	South Asian – 3.1%	No – 77.1%	A – 24.2%
Transgender female – 0.5%	3rd – 16.5%	Chinese – 6.3%	Not sure – 4.0%	– 15.5%
Transgender male – 0%	4th – 9.0%	Black – 2.4%		– 13.0%
Gender variant or non-conforming –	5th – 2.7%	Filipino – 0.5%		B – 10.7%

0.9%			
		Latin	
Other – 0.2%	6th – 1.5%	American – 1.2%	B – 7.7%
	Other – 0%	Arab – 6.8%	– 5.5%
		Southeast Asian – 3.6%	C – 3.0%
		West Asian – 0.7%	– 2.0%
		Korean – 0.5%	D – 1.7%
		Japanese – 0.5%	F – 0.5%
		Multiple visible minorities – 0.5%	
		Not a visible minority – 72.2%	
		Other – 1.7%	

^a Self reported.

^b Only requested in March 2019, September 2019, and November 2019 surveys.

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